

Software Engineering Take Home Test

Important Notes

- Please follow all the instructions and requirements provided in this document.
- You can make assumptions to any information not specified.
- You can use any AI coding assistant or tool to do this test but you are expected to check the correctness of the solution and be familiar with the code.
- Use Git to manage and track changes of your source code.
- Please complete the test and submit the required deliverables by email within 3 calendar days upon receiving the test. If you are unable to complete within 3 days, you can inform us by email that you require up to 2 more days.
- If you have any query, reach out to us via email:
 - Paul Weng: paul_weng@htx.gov.sg
 - Alvin Koh: alvin_koh@htx.gov.sg
 - Benjamin Ng: benjamin_ng@htx.gov.sg
 - Vincent Goh: vincent_goh@htx.gov.sg

Background

As a full-stack software engineer in the xDigital AI Products Team, you will be responsible for developing web applications integrated with AI capabilities. You are expected to adopt software engineering best practices and ensure high code quality. You need to have a good understanding of frontend technologies (such as Javascript/Typescript and React), backend technologies (such as Javascript/Typescript, Node.js, Python), database services (such as PostgreSQL) and GenAI capabilities (such as LLM).

Overview

In this test, you are required to develop both the frontend and backend of a **Task Assignment** application.

- 1) Frontend developed using **Typescript, React or a framework using React**.
- 2) Backend developed using **Typescript, Node.js or a framework using Node.js**.
- 3) Database using **PostgreSQL**.

You may use additional code libraries or dependencies as necessary.

Part 1: Database Design & Setup

1.1 Create a PostgreSQL database that supports the following entities and business logic.

Entities

- Developers
 - A Developer has a name (e.g. Alice, Bob).
 - A Developer can be assigned Task(s).
 - A Developer can have Skill(s) (e.g. Frontend, Backend, or both).
- Tasks
 - A Task has a title (e.g. “As a visitor, I want to see a responsive homepage so that I can easily navigate on both desktop and mobile devices.”)
 - A Task can require specific Skill(s) (e.g. Frontend, Backend, or both).
 - A Task can only be assigned to a Developer with the Skill(s) required by the Task.
 - A Task can have status “To-do”, “Done”, etc.
- Skills
 - A Skill is not unique to a Developer. Multiple Developers can have the same Skill.

1.2 Seed the database with the following data for Developers and Skills.

Developer	Skills
Alice	Frontend
Bob	Backend
Carol	Frontend, Backend
Dave	Backend

Part 2: Backend API Implementation

2.1 Create backend API using Typescript, Node.js or a framework using Node.js for the following operations.

Operations

- Tasks
 - Create
 - A new Task can be created.
 - Read
 - A Task and all its relevant properties can be read.
 - Update
 - A Task can only be assigned to a Developer with the Skill(s) required by the Task.
 - A Task can have its status changed.
- Developers
 - Read
 - A Developer and all their relevant properties can be read.
- Skills
 - Read
 - A Skill and all its relevant properties can be read.

Part 3: Frontend Implementation

3.1 Create web frontend for a single-page application (SPA) using Typescript, React or a framework using React for the following pages.

Pages

- Task List Page
 - All the existing Tasks and their relevant attributes should be listed.
 - A Task can be assigned to a Developer with the required Skill(s).
 - The status of a Task can be updated.
 - A simple wireframe is provided below as reference.

Tasks				
Task Title	Skills	...	Status	Assignee
As a visitor, I want to see a responsive homepage so that I can easily navigate on both desktop and mobile devices.	Frontend		To-do ▼	Alice ▼
As a system administrator, I want audit logs of all data access and modifications so that I can ensure compliance with data protection regulations and investigate any security incidents.	Backend		To-do ▼	▼
As a logged-in user, I want to update my profile information and upload a profile picture so that my account details are accurate and personalized.	Frontend, Backend		To-do ▼	▼

- Task Creation Page
 - A new Task can be created by submitting a form.
 - At the point of creating new Task, user can specify Skill(s) required by the Task.
 - At the point of creating new Task, user does not need to assign a Developer.
 - A simple wireframe is provided below as reference.

Create Task

New Task Component

Save

Part 4: Subtask Feature

- 4.1 There is a feature requirement that a Task can have subtask(s) and each subtask has the same properties as a Task. Modify the database schema to support this.
- 4.2 A Task can only have its status changed to “Done” if all its subtasks have their status as “Done”. Modify the solution to support this business logic.
- 4.3 Modify the frontend such that the Task creation page supports the creation of subtasks and nested subtasks. Additional React components should be created and rendered dynamically on the same page for these subtasks. A simple wireframe is provided below as reference for this.

The wireframe shows a form titled "Create Task(s)". It contains four nested subtask components, each represented by a dashed rectangular box. The first box contains the text "New Task Component 1" and an "Add Subtask" button. The second box, nested under the first, contains "New Task Component 1.1" and an "Add Subtask" button. The third box, nested under the second, contains "New Task Component 1.1.1" and an "Add Subtask" button. The fourth box, nested under the third, contains "New Task Component 1.1.2" and an "Add Subtask" button. At the bottom right of the form is a "Save" button.

Part 5: Skill Identification Using LLM

- 5.1 For Tasks and subtasks without specified Skill(s), there is a feature requirement to use LLM to identify the required Skill(s) based on the Task title.

For example:

“As a visitor, I want to see a responsive homepage so that I can easily navigate on both

desktop and mobile devices.” → [Frontend]

“As a system administrator, I want audit logs of all data access and modifications so that I can ensure compliance with data protection regulations and investigate any security incidents.” → [Backend]

“As a logged-in user, I want to update my profile information and upload a profile picture so that my account details are accurate and personalized.” → [Frontend, Backend]

- 5.2 Integrate your backend with a LLM using API to achieve the above. Feel free to use any LLM provider such as Gemini, which offers free access of their API.
- 5.3 There is no need for the user to manually trigger this process on the frontend. Whenever a new Task (including subtask) without user-specified Skill(s) is created, required Skill(s) will be automatically generated using LLM on the backend.

Part 6: Containerize Solution

- 6.1 Containerize your entire solution, including all dependencies, using Docker.
- 6.2 Ensure that your container(s) includes a fully operational application that can be executed using docker-compose.

Part 7: Submit Deliverables

- 7.1 Push your source code to a remote code repository like Github and make it publicly accessible. There should be a README file in your source code to document how to configure and run your application. Any documentation on system design, API and justification of the use of code libraries and dependencies should be included in the README file as well.
- 7.2 Submit your deliverables by emailing the link of your code repository and all necessary information (such as API keys) to the following:
 - Paul Weng: paul_weng@htx.gov.sg
 - Alvin Koh: alvin_koh@htx.gov.sg
 - Benjamin Ng: benjamin_ng@htx.gov.sg
 - Vincent Goh: vincent_goh@htx.gov.sg